

Referee Report: “When Volatility Targeting Crowds: An Agent-Based Tipping Point Analysis”

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Summary Assessment

Category	Count
HIGH severity issues	5
MEDIUM severity issues	8
LOW severity issues	6
Citation errors	3
Total issues	22

Overall recommendation: Major Revision. The paper addresses a timely and important question—at what adoption level does volatility targeting become self-defeating?—and proposes a novel agent-based simulation framework. However, several high-severity issues in model calibration, robustness, and overstated claims must be resolved before publication. The citation list contains factual errors that must be corrected.

1 Logic and Structure

1. **[HIGH] Missing sensitivity analysis for key parameters.** The results depend critically on $\lambda = 0.005$ (Kyle lambda), $\gamma = 200$ (VIX amplification), and $\kappa = 0.03$ (mean-reversion speed), yet no sensitivity analysis is provided. The tipping point between 30–50% could shift dramatically with different λ values. Without showing how the threshold responds to parameter perturbation (± 20 –50%), the central claim (“tipping point at 30–50%”) lacks robustness.

Recommendation: Add a full sensitivity analysis table varying $\lambda \in \{0.002, 0.005, 0.010\}$, $\gamma \in \{100, 200, 400\}$, and $N \in \{500, 1000, 5000\}$. Report how the tipping point shifts across these combinations.

2. **[MEDIUM] Logical gap between model mechanism and real-world VT.** The model assumes all VT agents use the identical 12/VIX rule. In practice, VT strategies use heterogeneous estimators (GARCH, EWMA, realized vol) with different lookback windows. The paper acknowledges this in Limitations but does not discuss the *direction* of the bias: does homogeneity overstate or understate the tipping point? The Limitations section (Section 4.3) states it “likely smooth[s] the tipping point,” but this claim itself needs justification or simulation evidence.
3. **[MEDIUM] No equilibrium analysis.** The introduction draws a parallel to the Brunnermeier–Pedersen (2009) liquidity spiral, but that model has an equilibrium foundation. The ABM here is purely mechanical—there is no optimization, no rational expectations, no equilibrium price. This limits the theoretical contribution. The paper should explicitly position itself relative to equilibrium models and clarify what the ABM adds that equilibrium analysis cannot.
4. **[LOW] Section structure for FRL.** Finance Research Letters has a strict page limit (typically 5–7 pages in final format). The current manuscript, with double spacing, is approximately 8–9 printed pages. The Discussion section (Section 4) could be compressed; the Limitations subsection is admirably thorough but could be shortened by moving some items to footnotes.

2 Research Motivation and Literature Gap

5. **[HIGH] Literature gap claim is overstated.** The paper claims to provide “the first quantitative crowding threshold for VT strategies,” but several related works are not cited:
 - **Cejnek et al. (2021)**, “The Popularity Asset Pricing Model,” *European Financial Management*—develops a formal framework for how strategy popularity affects returns, directly relevant to crowding.
 - **Falconio (2023)**, “Carry, factor investing, and crowding,” *Journal of Banking & Finance*—quantifies crowding thresholds in carry strategies.
 - **Hasse and Lavin (2022)**, “Systemic risk and procyclicality of volatility targeting strategies”—directly addresses VT procyclicality with a formal model.
 - **Moallemi and Saglam (2013)**, “The cost of latency in high-frequency trading”—relevant to the Kyle market-maker setup.

Without engaging with this literature, the gap claim is weakened.

Recommendation: Expand the literature review to include these references and clarify the marginal contribution relative to existing crowding frameworks.

6. **[MEDIUM] Motivation conflates AUM with adoption fraction.** The introduction cites “USD 2 trillion in short-volatility and volatility-sensitive strategies” (Bouchaud et al., 2018), but this AUM figure includes very different strategies (short vol, risk parity, vol-of-vol trading). The paper’s adoption fraction ϕ refers specifically to 12/VIX-type VT, which is a subset. The disconnect between the motivating statistic and the model variable weakens the real-world grounding.

Recommendation: Provide a more targeted AUM estimate for VT strategies specifically, or clearly acknowledge the discrepancy and explain why ϕ is a meaningful abstraction despite this gap.

7. **[LOW] The “currently below 5%” claim needs a source.** The abstract and Discussion state that “Current real-world VT adoption is estimated below 5%,” but no citation or methodology is provided for this estimate. This is a key claim for the paper’s policy relevance.

3 Model Specification

8. **[HIGH] Kyle lambda is constant and exogenous.** In Kyle (1985), λ is *endogenous*—it depends on the variance of informed order flow and the informativeness of prices. Here $\lambda = 0.005$ is a fixed constant, which removes the key feature of the Kyle model: endogenous liquidity. When VT agents all sell simultaneously, a real market maker would widen spreads (increase λ), creating a nonlinear amplification that is *absent* from this model. Paradoxically, the paper claims to find a positive feedback loop, but the constant λ actually *dampens* the feedback relative to a realistic specification.

Recommendation: Either (a) make λ endogenous (e.g., $\lambda_t = \lambda_0 + \alpha \cdot |\text{NOF}_t/N|$, capturing reduced liquidity during one-sided flow), or (b) clearly acknowledge that constant λ is conservative and likely *understates* the tipping point severity.

9. **[HIGH] VIX dynamics lack option-market foundations.** The VIX is defined as $\sqrt{\frac{1}{T} \sum E[r^2]}$ from the options market, but the model’s VIX (Eq. 2) is a mean-reverting process driven by realized volatility deviations. This is closer to a realized-vol tracker than an implied-vol measure. The real VIX often diverges from realized vol (the “variance risk premium”), and this divergence is precisely what makes VT strategies profitable. By equating VIX with realized vol, the model may *overstate* the feedback loop (because VIX responds too mechanically to returns).

Recommendation: At minimum, calibrate the VIX process against the empirical relationship between realized vol and VIX. Ideally, introduce a variance risk premium component.

10. **[MEDIUM] Agent heterogeneity within VT class is zero.** All VT agents use the same 12/VIX rule, the same rebalancing frequency (daily), and the same leverage cap (1.5). Real VT implementations vary in: volatility estimator (GARCH vs. EWMA vs. realized vol), target level (σ^*), rebalancing frequency (daily/weekly/monthly), and position limits. This homogeneity mechanically maximizes the correlation of VT order flow, which is the key driver of the results.

Recommendation: Add a robustness check with heterogeneous VT agents (e.g., different target volatilities drawn from a distribution, different lookback windows).

11. **[MEDIUM] Noise trader specification is ad hoc.** Noise traders adjust weights by $\Delta w \sim N(0, 0.02)$, clipped to $[0, 1.5]$. This is a random walk bounded by clipping, which will tend to concentrate noise trader weights near the boundaries over time (boundary accumulation). This could interact with the VT feedback mechanism in non-obvious ways.

Recommendation: Consider an Ornstein–Uhlenbeck process for noise trader weights (mean-reverting to 0.5 or 1.0) to avoid boundary effects.

12. **[LOW] No transaction costs or slippage.** The absence of transaction costs means VT agents can rebalance frictionlessly every day. In practice, the cost of daily rebalancing is non-trivial and would reduce the incentive to follow VT precisely, acting as a natural dampener on crowding. This likely overstates the severity of the tipping point.

4 Results and Statistical Reliability

13. **[HIGH] 100 simulations may be insufficient for tail statistics.** The paper reports kurtosis (253.8 at 100% adoption) and flash crash frequency based on 100 simulations. For heavy-tailed distributions, 100 draws provide very imprecise estimates of the fourth moment. The standard error of the sample kurtosis for a distribution with true kurtosis κ is approximately $\sqrt{24/n} \approx 0.49$ for $n = 100$ under normality, but for heavy-tailed distributions, the SE explodes. The reported kurtosis of 253.8 at 100% adoption has an enormous confidence interval.

Recommendation: Either (a) increase to $M = 1,000$ simulations, or (b) report bootstrap confidence intervals for all metrics, especially kurtosis and flash crash frequency. At minimum, report standard errors across the 100 simulations.

14. **[MEDIUM] No confidence intervals or standard errors in tables.** Tables 1 and 2 report only means across 100 simulations. Without standard errors or confidence intervals, the reader cannot assess whether the differences between adjacent

adoption levels (e.g., 20% vs. 30%) are statistically significant. The t -test in Section 3.3 partially addresses this, but only for selected comparisons.

Recommendation: Add standard errors in parentheses for all entries in Tables 1 and 2. Report the full pairwise t -test matrix, not just selected comparisons.

15. **[MEDIUM] The “phase transition” language is imprecise.** The paper describes the performance degradation as a “phase transition” (p. 5), but this term has a specific meaning in statistical physics: a discontinuity in an order parameter at a critical point. The observed pattern—gradual degradation followed by sharp decline—is better described as a *nonlinear threshold effect* or *tipping point*, not a phase transition. A true phase transition would require analytical evidence of a bifurcation in the model’s dynamics.
16. **[LOW] Flash crash frequency artifact at 100%.** Table 1 notes that the flash crash frequency *declines* at 100% adoption, attributed to inflated σ (raising the 3σ threshold). While the footnote explanation is correct, this metric is therefore uninformative at extreme adoption levels. Consider also reporting the frequency of absolute returns $> x\%$ for a fixed threshold (e.g., 5% or 10% daily moves) that does not depend on the sample standard deviation.
17. **[LOW] Correlation between VIX changes and returns.** Section 3.4 reports the contemporaneous correlation changes from “near zero at 0% to -0.06 at 100%.” A correlation of -0.06 is remarkably small in absolute terms, even if statistically significant with 252,000 observations. The empirical VIX–return correlation is typically around -0.7 to -0.8 . This suggests the model’s VIX dynamics are severely miscalibrated.

Recommendation: Report the empirical VIX–return correlation and compare it to the model’s. If the model cannot reproduce this basic stylized fact, it raises concerns about the external validity of the tipping point estimate.

5 Discussion: Policy Implications and Overclaiming

18. **[MEDIUM] Policy recommendation exceeds the evidence.** Section 4.1 recommends that the “European Systemic Risk Board and the Financial Stability Oversight Council could incorporate VT adoption metrics into their macroprudential surveillance frameworks.” This is a strong policy recommendation based on a stylized simulation with known limitations (constant λ , homogeneous agents, no transaction costs, simplified VIX). The qualification “preliminary simulation results” appears in the abstract but is not adequately carried through to the policy discussion.

Recommendation: Either strengthen the simulation (endogenous λ , heterogeneous agents, calibration to data) to support such recommendations, or significantly temper the policy language to “our results *motivate further investigation* into whether VT adoption monitoring would be useful.”

19. **[LOW] No comparison to portfolio insurance literature.** The paper mentions the 1987 crash (Genotte and Leland, 1990) in passing, but does not systematically compare VT crowding to portfolio insurance crowding. The mechanisms are closely related (both involve procyclical selling), and the portfolio insurance literature provides empirical estimates of the adoption level at which destabilization occurred. This would strengthen the paper’s contribution.

6 Limitations Assessment

20. **[LOW] Limitations section is thorough but could go further.** The five limitations listed are appropriate. Two additional limitations should be acknowledged:
 - **Single asset:** The model has only one risky asset. In practice, VT rebalancing involves selling equities and buying bonds/cash, and the cross-asset dynamics (flight to quality, bond rally during equity selloffs) would moderate the feedback.
 - **No central bank or circuit breaker:** Real markets have circuit breakers, central bank interventions, and other stabilization mechanisms that would truncate the feedback loop. Their absence biases the model toward more extreme outcomes at high ϕ .

7 Citation Verification

Each bibliography entry was verified against publisher databases. Results are summarized below.

Citation	Status	Details
Moreira and Muir (2017)	CORRECT	<i>Journal of Finance</i> , 72(4), 1611–1644. Verified.
Harvey et al. (2018)	CORRECT	<i>Journal of Portfolio Management</i> , 45(1), 14–33. Published Fall 2018, issue dated October 2018. Verified.

Citation	Status	Details
Bouchaud et al. (2018)	CORRECT	Cambridge University Press, 2018. Verified. However, the citation is used to support the claim that “short-volatility and volatility-sensitive strategies collectively manage over USD 2 trillion.” This specific AUM claim should be verified against the exact passage in the book; it may originate from the ECB (2020) report instead.
Brunnermeier and Pedersen (2009)	CORRECT	<i>Review of Financial Studies</i> , 22(6), 2201–2238. Verified.
ECB (2020)	ISSUE	The bibliography says “ <i>Financial Stability Review</i> , November.” The VT-specific analysis (“Volatility-targeting strategies and the market sell-off”) appeared in the May 2020 issue, not November. The November 2020 FSR exists but does not contain the specific VT box. Should be corrected to “May 2020.”
Gennotte and Leland (1990)	CORRECT	<i>American Economic Review</i> , 80(5), 999–1021. Verified.
Harvey (2016)	ERROR	The bibliography lists year as 2016, and the text cites “Harvey (2016).” The paper was published in the <i>Journal of Finance</i> , 72(4), 1399–1440, in August 2017 , not 2016. The presidential address was <i>delivered</i> in 2016, but the published article is Harvey (2017). Must be corrected to 2017.
Kyle (1985)	CORRECT	<i>Econometrica</i> , 53(6), 1315–1335. Verified.
Baltas (2019)	CORRECT	<i>Financial Analysts Journal</i> , 75(3), 89–104. Verified.

Citation	Status	Details
Bookstaber et al. (2014)	ERROR	The bibliography cites 2014 and lists volume 13, pp. 433–466 in the <i>Journal of Economic Interaction and Coordination</i> . The OFR working paper was issued in 2014, but the journal publication was in 2018 (published online February 2017, print 2018, Vol. 13, pp. 433–466). The bibitem year should be 2018, not 2014. If citing the working paper, the journal/volume/pages should be removed.
Perchet et al. (2016)	ISSUE	The title listed is “Inter- and intra-asset diversification in risk-based portfolios with the 1/VIX rule,” <i>Journal of Portfolio Management</i> , 42(3), 84–98. This exact title and venue could not be verified. The same authors published “Predicting the Success of Volatility Targeting Strategies: Application to Equities and Other Asset Classes” in the <i>Journal of Alternative Investments</i> , 18(3), 21–38 (2015), and “Inter-Temporal Risk Parity” on SSRN (2014). No exact match for the cited title/journal/volume was found in SSRN, Google Scholar, or pm-research.com. This citation requires verification of the exact title and venue.

Citation error summary:

- **2 factual errors:** Harvey (2016→2017), Bookstaber et al. (2014→2018)
- **2 issues:** ECB (November→May 2020), Perchet et al. (title/venue unverifiable)

8 Specific Revision Recommendations

Priority 1 (Must fix before resubmission)

1. Add sensitivity analysis for λ , γ , κ , and N (Section 3, Issue 1).
2. Either make λ endogenous or explicitly justify constant λ as conservative (Issue 8).
3. Increase simulations to $M \geq 500$ or provide bootstrap CIs for all tail statistics (Issue 13).
4. Add standard errors to all table entries (Issue 14).
5. Correct all citation errors: Harvey 2016→2017, Bookstaber 2014→2018, ECB November→May, verify Perchet et al. (Section 7).
6. Expand literature review: Cejnek et al. (2021), Falconio (2023), Hasse & Lavin (2022) (Issue 5).

Priority 2 (Strongly recommended)

7. Add robustness check with heterogeneous VT agents (different σ^* , lookbacks) (Issue 10).
8. Calibrate VIX–return correlation against empirical data and report comparison (Issue 17).
9. Temper policy recommendations to match the evidence level (Issue 18).
10. Report absolute-threshold flash crash frequency (e.g., $|r| > 5\%$) alongside 3σ (Issue 16).

Priority 3 (Nice to have)

11. Shorten paper for FRL format constraints (Issue 4).
12. Add mean-reverting noise trader specification (Issue 11).
13. Discuss single-asset vs. multi-asset bias (Issue 20).
14. Compare to portfolio insurance crowding thresholds from 1987 literature (Issue 19).
15. Provide source for “below 5% adoption” claim (Issue 7).

Conclusion

This paper tackles an important and timely question at the intersection of dynamic portfolio management and systemic risk. The agent-based approach is appropriate for studying emergent crowding phenomena that equilibrium models struggle to capture. The nonlinear tipping point finding is the paper's key contribution and is intriguing.

However, the central quantitative claim (30–50% threshold) cannot be accepted without sensitivity analysis showing its robustness to parameter choices. The constant Kyle lambda, homogeneous VT agents, and absence of transaction costs all bias the model in ways that are difficult to sign without further analysis. The citation list requires correction. With these revisions, the paper could make a valuable contribution to the literature on strategy crowding and systemic risk.

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